Backing Up the SSD System

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During development, users may need to back up the system to prevent subsequent development from affecting the current system environment.

```
The image file demonstrated in the tutorial is not the actual factory image; it
is only for tutorial demonstration
```

1. Hardware Connection

Users need to prepare an SSD enclosure in advance, install the SSD into the SSD enclosure, and connect it to the computer or virtual machine: the computer and the virtual machine system need to be Ubuntu systems.



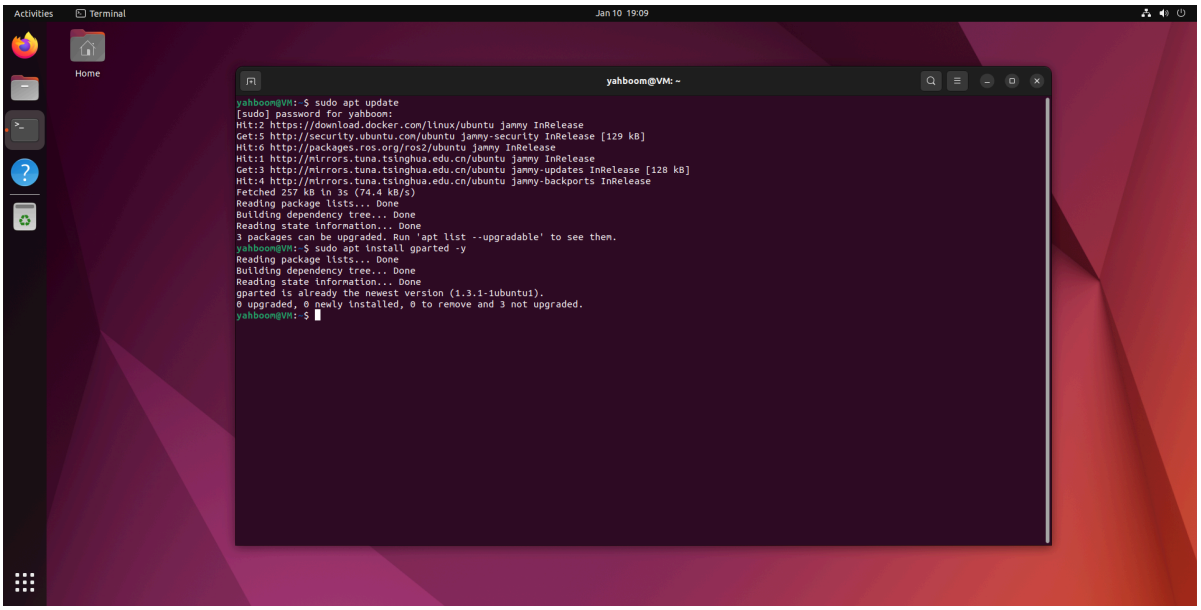


2. Compressing the SSD

Since the SSD capacity provided with the Jetson Orin series mainboard is relatively large, it needs to be compressed to an appropriate space for system backup, which saves time for backing up and writing the system.

2.1. Installing Gparted

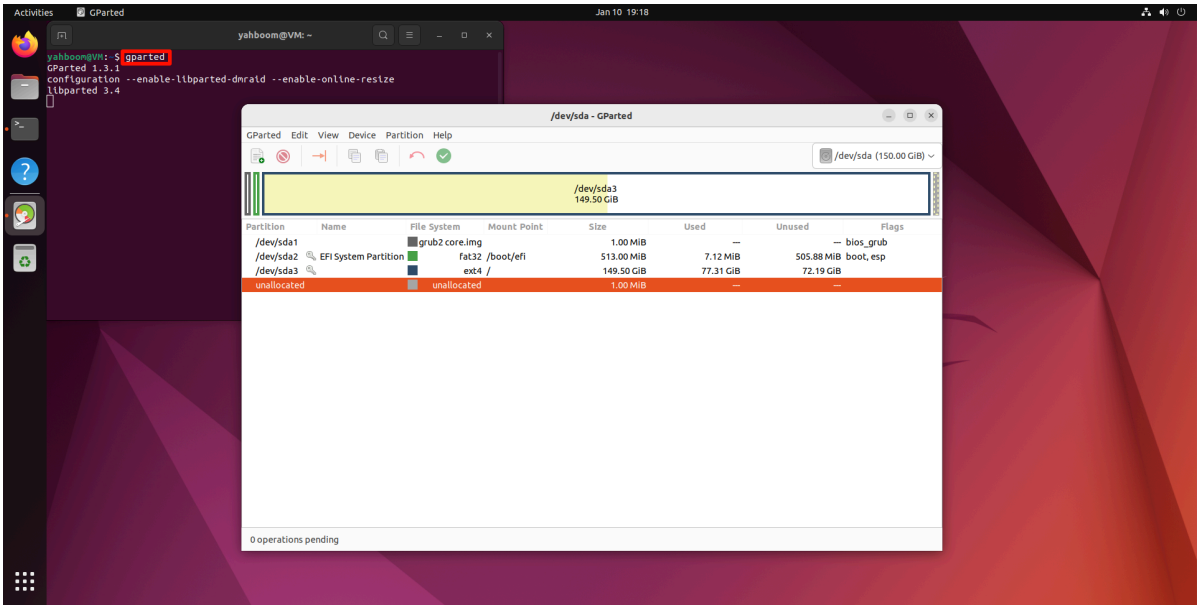
```
sudo apt update
sudo apt install gparted -y
```



2.2. Using GParted

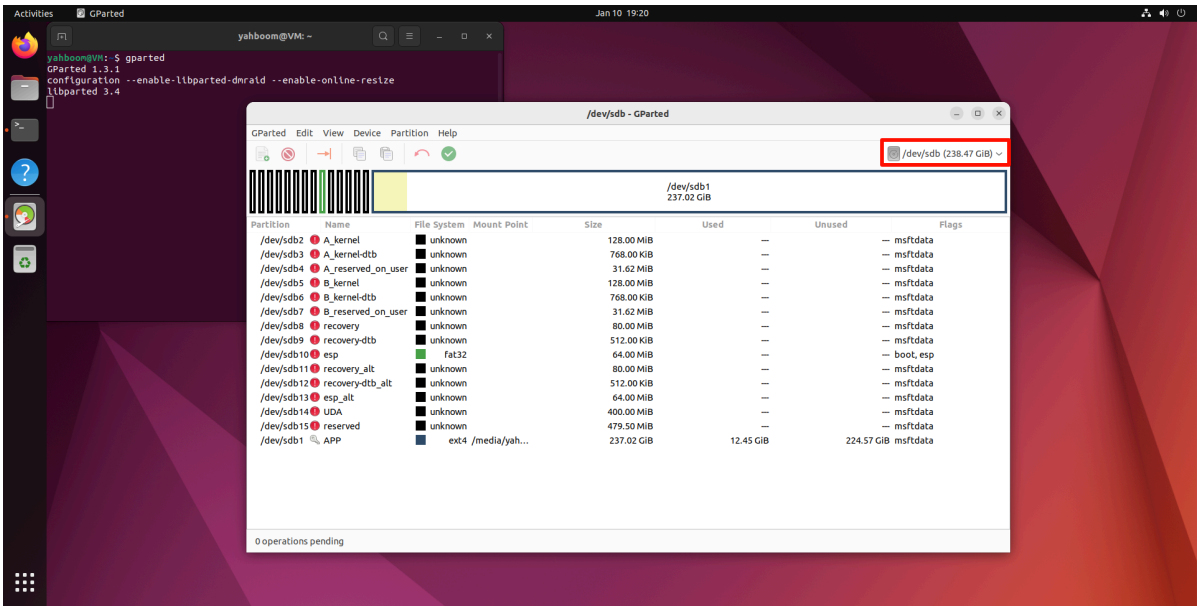
Find the `GParted` application icon in the system application menu bar to open it, or enter the command below in the terminal to start it:

```
gparted
```



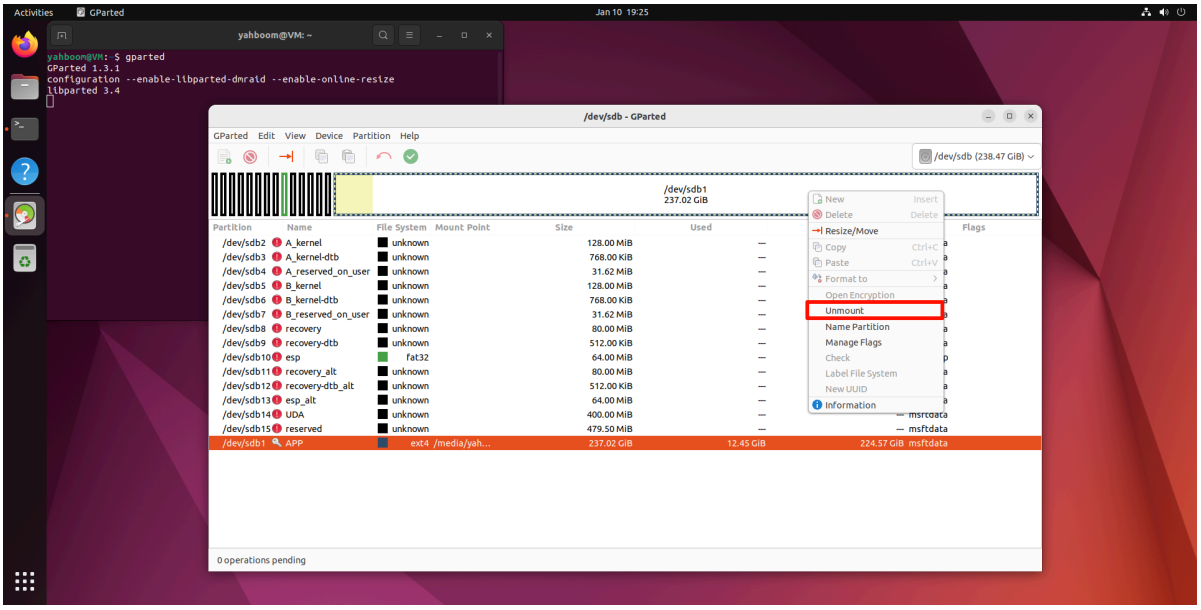
2.2.1. Selecting the SSD

Select the newly added disk symbol: you can confirm again whether it is the SSD you mounted according to the disk capacity



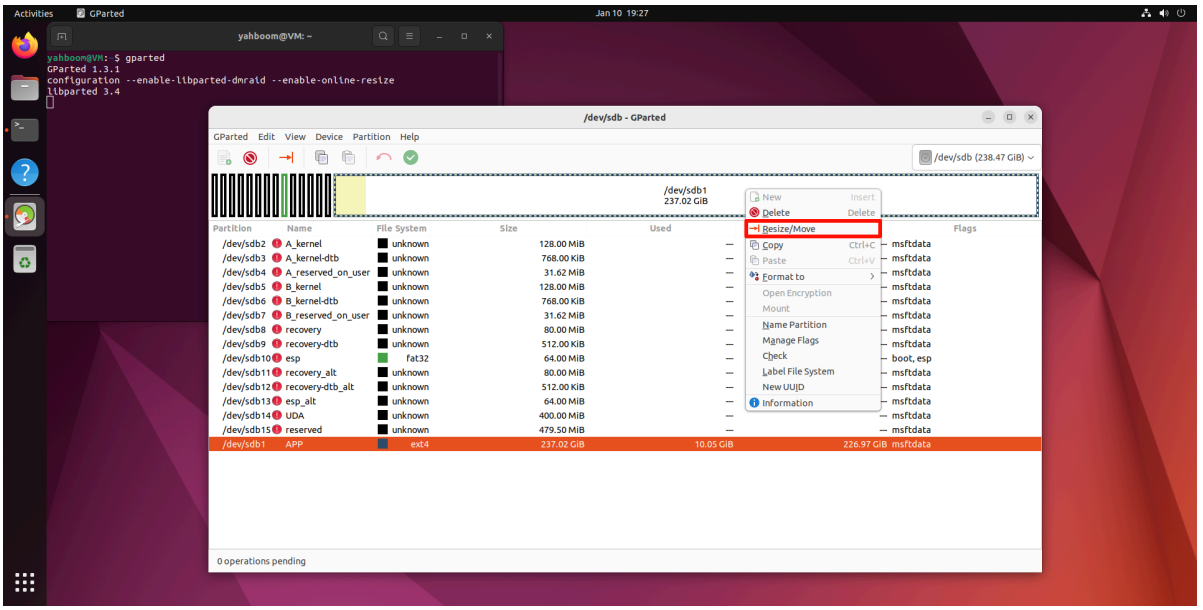
2.2.2. Unmounting the Partition

Before operating on the disk, the disk needs to be unmounted first: select the `APP` partition (the largest partition) on the disk, and click `Unmount` to unmount the partition

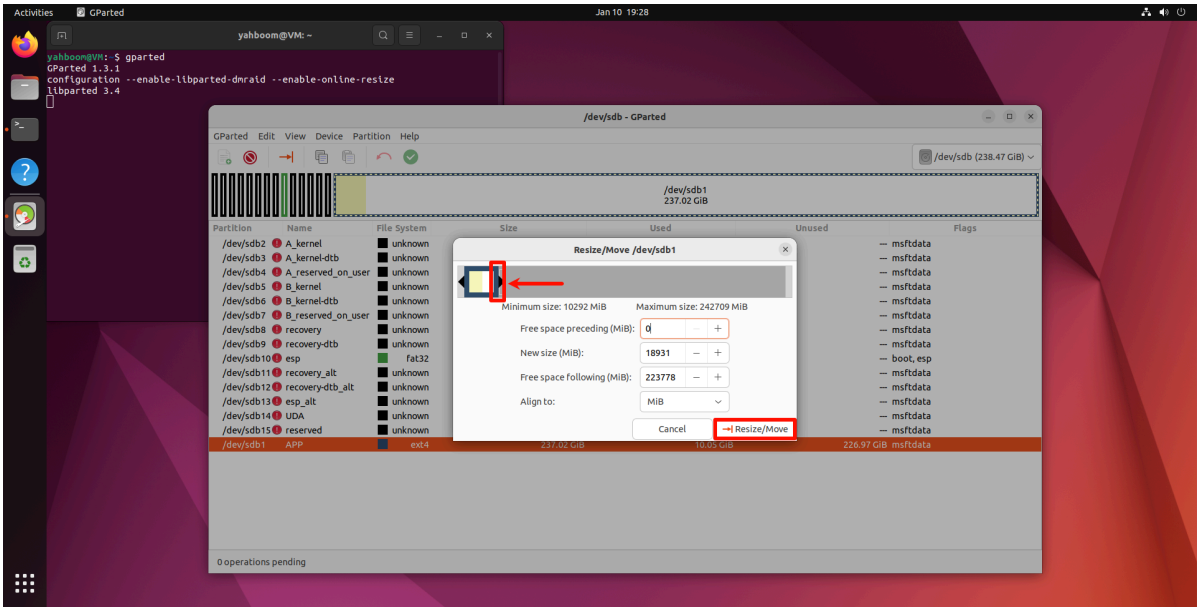


2.2.3. Compressing the Disk

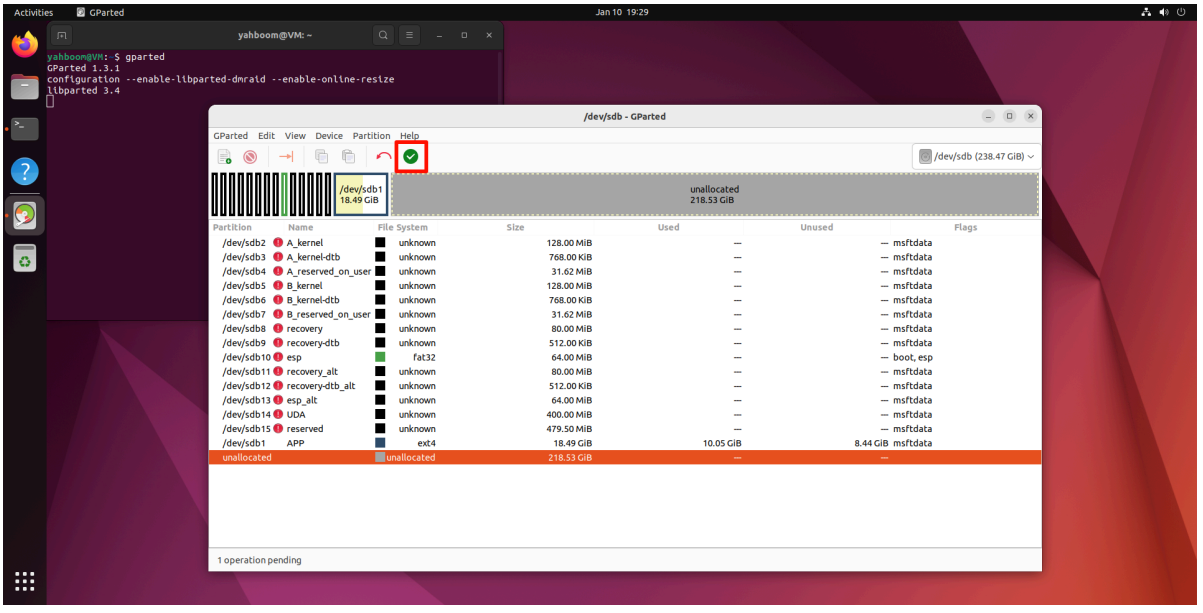
Right-click the unmounted disk partition and readjust the size of the partition space that was unmounted:

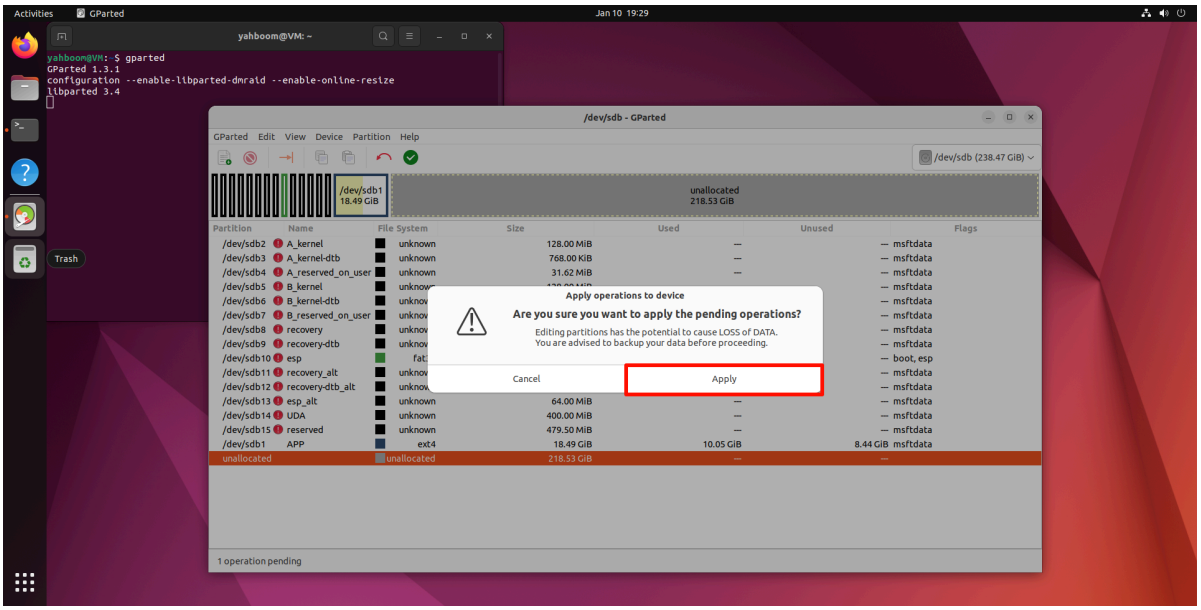


The partition size can be adjusted with the slider: yellow is the space used by the partition, and white is the unused space. It is recommended to leave about 5-10G of unused space in the partition to avoid the system failing to start

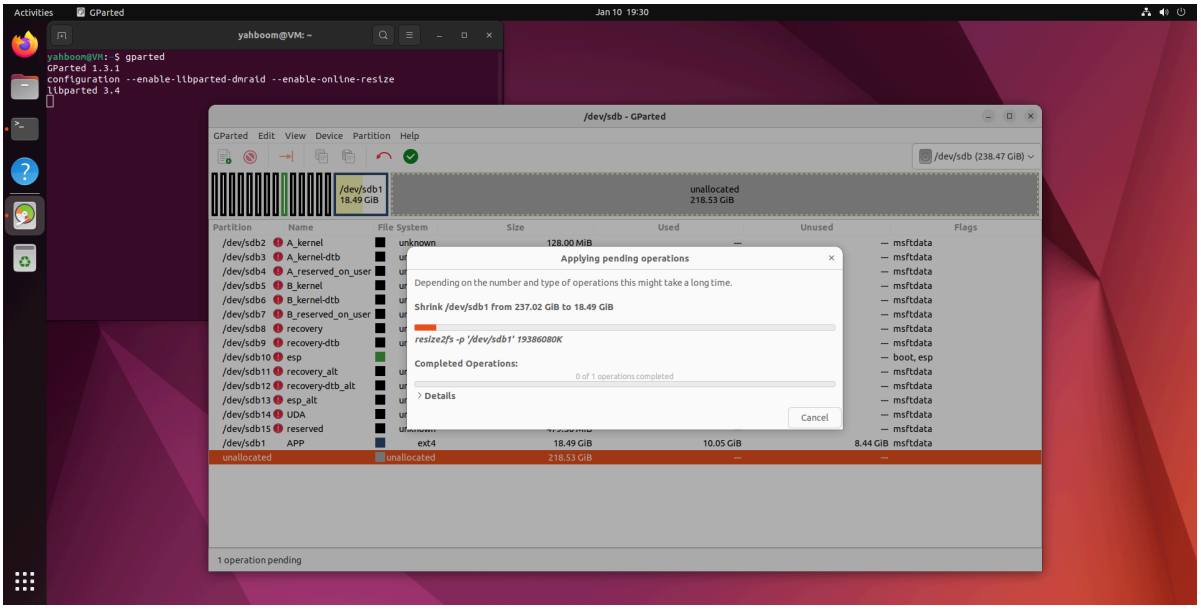


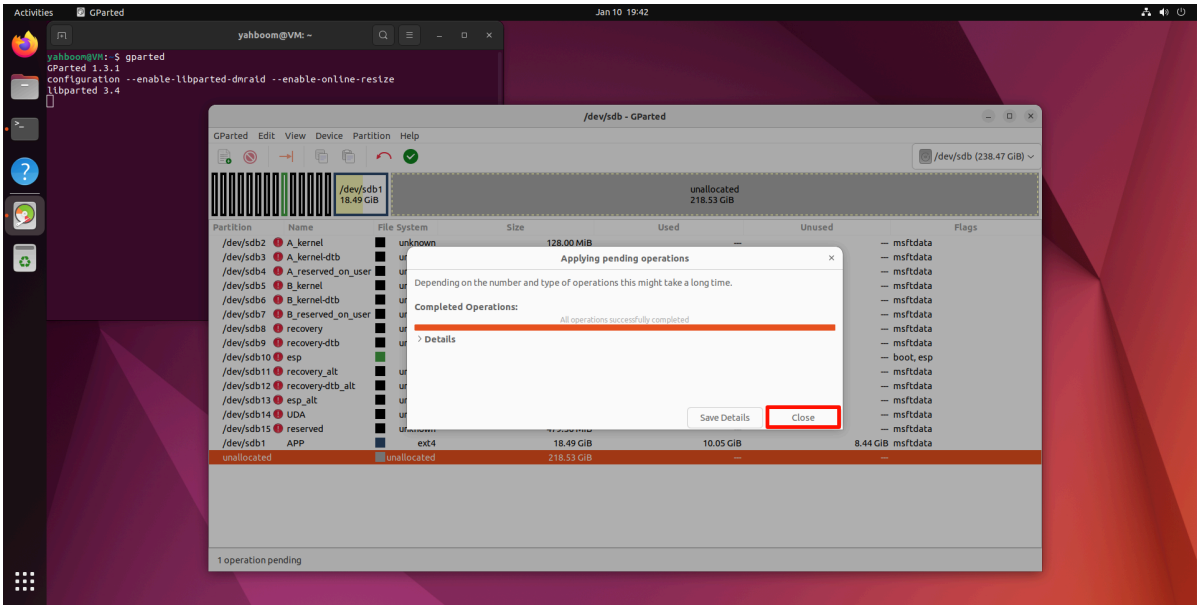
Confirm the disk operation:





Wait for the operation to complete:





Close GParted after completing the above operations!

3. Backing Up the SSD

3.1. Viewing Disk Information

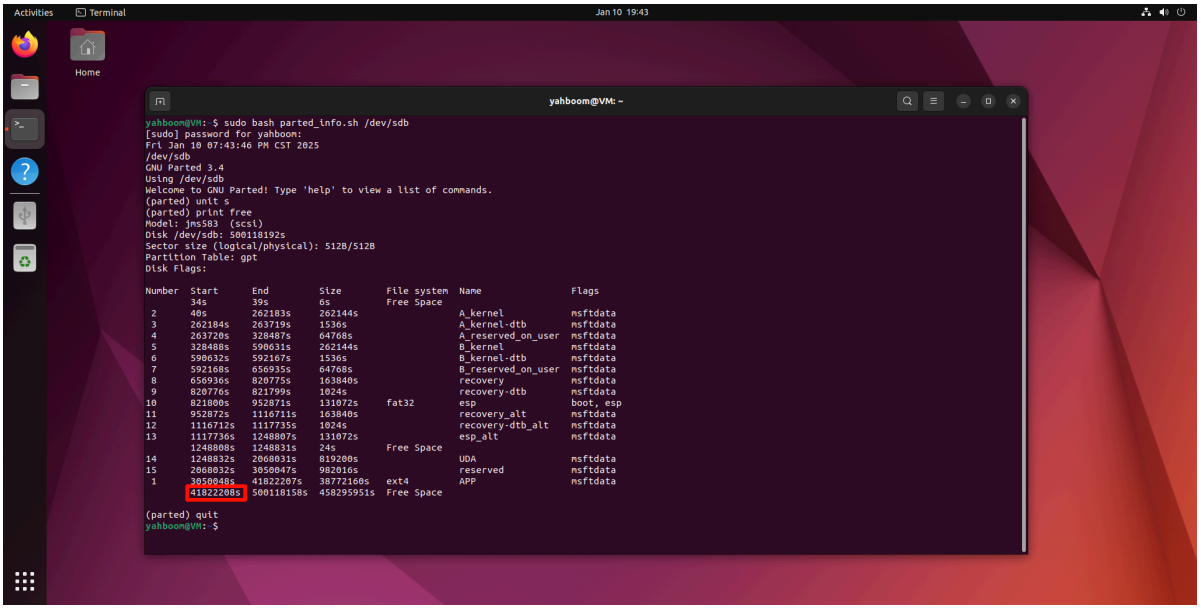
Open the terminal and use the script to view the current disk information: the drive letter needs to correspond to the drive letter of the SSD you are backing up

```
sudo bash parted_info.sh /dev/sdb
```

> **parted_info.sh Script Content**

```
#!/bin/bash
date
echo $1
sudo parted $1 <<EOF
unit s
print free
quit
EOF
```

Record the data in the figure: 41822208s



3.2. Starting the Disk Backup

Use the `dd` command to back up the SSD to an `img` file. Enter the following in the terminal:

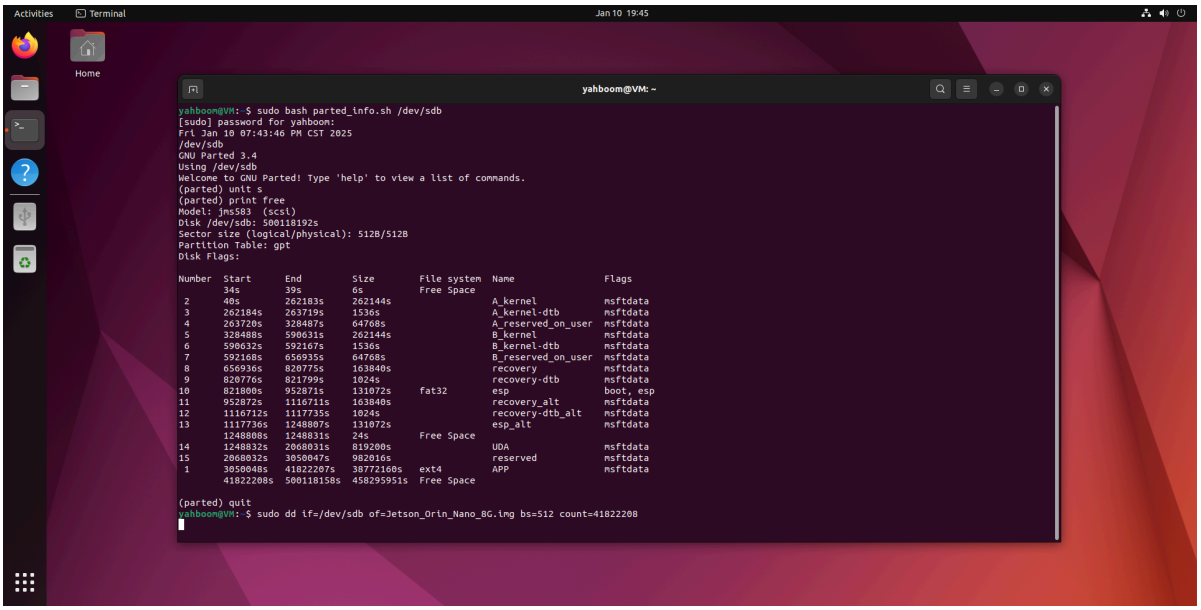
```
sudo dd if=/dev/sdb of=Jetson_Orin_Nano_8G.img bs=512 count=41822208
```

`/dev/sdb`: SSD drive letter

`Jetson_Orin_Nano_8G.img`: Image name

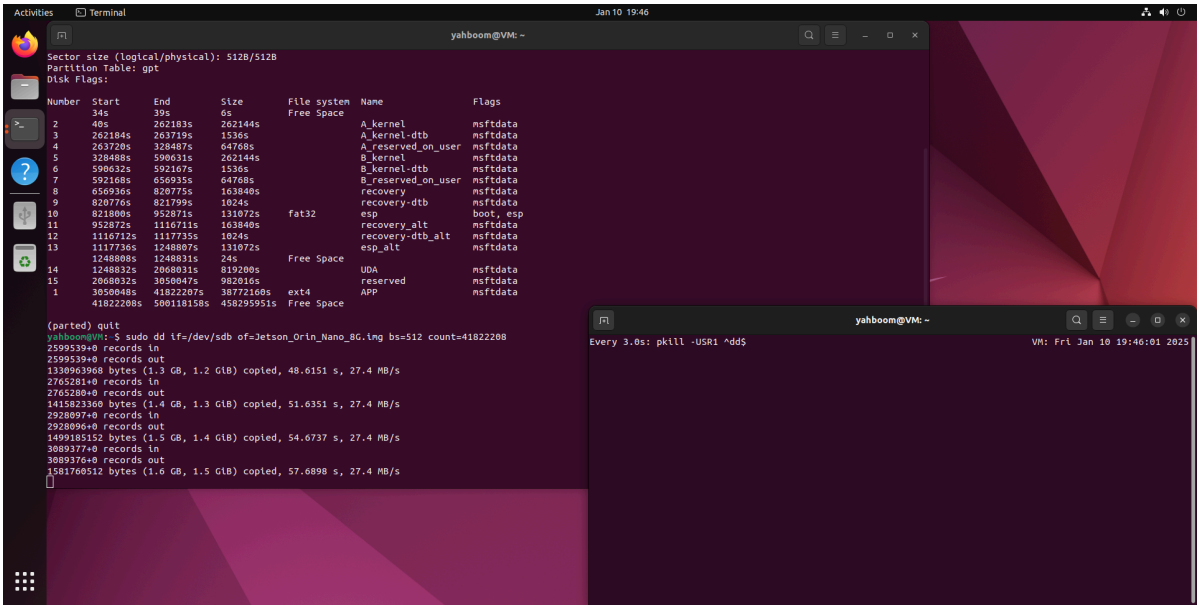
`bs=512`: Set the block size to 512 bytes

`41822208`: Data queried by the script

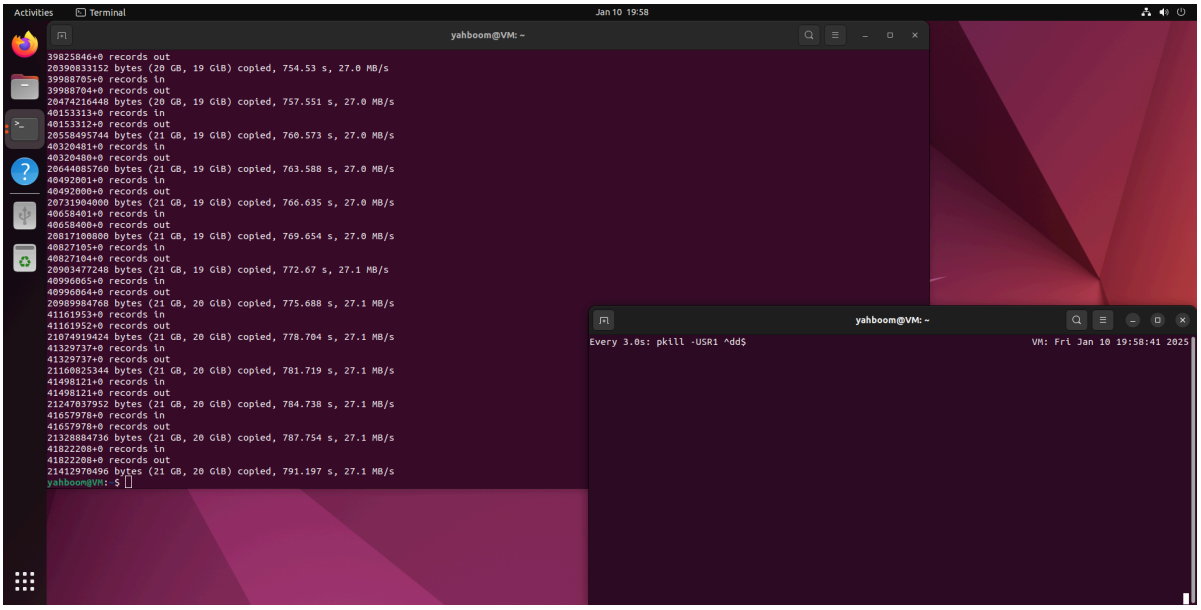


To view the `dd` process information, open another terminal and enter the command below:

```
sudo watch -n 3 pkill -USR1 ^dd$
```



Wait for the backup to complete:



After the system backup is complete, move the backup file (Jetson_Orin_Nano_8G.img) to the Windows system for use by yourself.

